

A singleton counterexample to conjecture 00000008471

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4 October 2026

The final clause asserts that the complement of the locked continuous observables is a dense open set. We disprove density under the explicit locking definition given in both versions of the conjecture.

Let $X = \{a\}$, with its unique compact topology, and let $T = \text{id}_X$. Its only Borel probability measure is δ_a : any probability measure assigns mass 1 to $X = \{a\}$, and these are all its measurable sets. This measure is invariant under T . For every continuous $f : X \rightarrow \mathbb{R}$ we have

$$\int_X f d\delta_a = f(a).$$

Since there is no other invariant probability measure, the actual maximizing-measure set is

$$M(f) = \{\delta_a\} \quad \text{for every } f \in C(X, \mathbb{R}).$$

This conclusion follows from maximization over all invariant probability measures, rather than from a restricted list of candidates.

The point a has period 1, and the equidistribution on its orbit is

$$\frac{1}{1} \sum_{j=0}^0 \delta_{T^j(a)} = \delta_a.$$

Consequently every continuous observable is locked in the stated sense: its entire maximizing-measure set consists of the equidistribution on one periodic orbit. In particular the locked-function set is the whole space $C(X, \mathbb{R})$. Its complement is empty. The function space is nonempty (it contains the zero function), so the empty set is not dense in its usual uniform topology. Thus the claimed dense open complement law is false. Although the empty complement is open, it fails the required density.

Neither source excludes singleton dynamical systems. Even if locking also requires persistence under perturbation, this example satisfies that condition: every continuous perturbation still has the same unique maximizing measure. The counterexample concerns the final conjunct; it does not assert a failure of the generic-locking conjunct.

Formal verification. The accompanying Lean project uses the actual compact space `Unit`, its Borel probability measures, the actual identity map, Bochner integrals and `ContinuousMap Unit Real`. It proves uniqueness of every probability measure, the full maximizing-measure characterization, the weighted orbit measure and periodicity, locking of every observable, and failure of Mathlib's actual `Dense` predicate for the complement. The project is pinned to Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`.